

RAJASRI K

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 4+ years of experience designing, developing, and deploying production AI and machine learning solutions across banking, healthcare, pharmaceuticals, and e-commerce. Experienced in building LLM-powered applications, RAG pipelines, and multi-agent AI systems using LangChain, LangGraph, OpenAI GPT models, and modern MLOps practices on AWS and GCP. Strong background in machine learning, NLP, and data engineering, delivering scalable AI solutions that automate business workflows, improve knowledge retrieval, and enable data-driven decision-making.

TECHNICAL SKILLS

Generative AI / LLMs: OpenAI GPT, Meta Llama, Amazon Bedrock, LangChain, LangGraph, MCP (Model Context Protocol), Multi-Agent Systems, Prompt Engineering, Function/Tool Calling, Fine-Tuning (LoRA, QLoRA, PEFT)

RAG/Vector Search: RAG, FAISS, Pinecone, ChromaDB, Embedding Models

Machine Learning: TensorFlow, PyTorch, XGBoost, ARIMA, Prophet, LSTM, Random Forest, Logistic Regression, Time Series Forecasting, Feature Engineering, Model Evaluation

NLP: Hugging Face Transformers, spaCy, Information Extraction, Document Intelligence

Languages & Data Engineering: Python, SQL, Java, PySpark, Databricks, Apache Spark

MLOps / Deployment: MLflow, Docker, Kubernetes, CI/CD, FastAPI

Cloud: AWS (SageMaker, Lambda, EC2, S3), GCP (Vertex AI, BigQuery, Cloud Storage, Pub/Sub)

Databases / BI / Tools: MySQL, Snowflake, Git, Power BI, Tableau, JIRA, SHAP

PROFESSIONAL EXPERIENCE

Generative AI Engineer

Regions Bank | Aug 2025 – Present

- Engineered production-grade RAG pipelines using FAISS, Pinecone, and embedding models to power semantic search across 100K+ banking policies and compliance documents, improving retrieval accuracy and reducing hallucinations by 30% through prompt optimization and evaluation.
- Designed agentic AI workflows with LangGraph, tool calling, and function calling to automate document review, case triage, and compliance support, reducing manual processing time by 45% while improving workflow efficiency.
- Productionized LLM and machine learning services using FastAPI, Docker, Kubernetes, MLflow, and CI/CD, implementing model versioning, monitoring, and drift detection to reduce deployment time and improve production reliability.
- Built scalable PySpark and SQL data pipelines processing 10M+ structured and semi-structured banking records to support RAG indexing, analytics, and AI model training, reducing ETL execution time.
- Partnered with compliance, risk, and business stakeholders to translate banking requirements into secure, production-ready GenAI solutions aligned with regulatory, governance, and responsible AI standards.
- Contributed to Agile development through sprint planning, peer code reviews, and technical documentation, improving collaboration and accelerating AI feature delivery.

Research Assistant

Texas Tech University | Jan 2025 – July 2025

- Designed a multi-agent research assistant using LangGraph, LangChain, and MCP, orchestrating Planner, Searcher, Retriever, Reader, Writer, Critic, and Reviewer agents to automate literature discovery, analysis, and report generation, reducing manual research effort by ~60%.
- Implemented RAG pipelines using FAISS, ChromaDB, and embedding models to enable semantic retrieval and context-aware synthesis of academic literature, improving the relevance and quality of generated research summaries.
- Investigated Mixture-of-Experts (MoE) architectures for multimodal learning by evaluating state-of-the-art foundation models, comparing routing mechanisms, training strategies, and performance trade-offs across different model designs.
- Evaluated modality fusion and adaptive expert-routing approaches for missing-modality scenarios to improve robustness and generalization in multimodal AI systems through experimental analysis.

- Collaborated with faculty researchers to analyze experimental results, document findings, and contribute to ongoing research in multimodal AI, agentic systems, and LLM-powered applications.

Data Scientist / Machine Learning Engineer

Torrent Pharmaceuticals | Ahmedabad, India | June 2023 – Jul 2024

- Engineered scalable Python, SQL, Java data pipelines processing 8M+ manufacturing, clinical trial, and supply chain records, improving data quality and reducing processing time by 35% to support analytics and machine learning initiatives.
- Developed and evaluated time-series forecasting models (ARIMA, Prophet, LSTM) for pharmaceutical demand planning, improving forecast accuracy by 15% and enabling more effective inventory optimization across manufacturing operations.
- Built NLP-based document intelligence solutions using spaCy, Hugging Face Transformers, and Vertex AI to automate information extraction and classification from regulatory documents and quality reports, reducing manual review effort and improving data accessibility.
- Performed feature engineering, model evaluation, and hyperparameter tuning to optimize machine learning model performance, leveraging cross-validation and business-driven evaluation metrics for robust predictions.
- Collaborated with cross-functional teams to translate manufacturing and business requirements into data-driven solutions, while developing Power BI and Tableau dashboards to communicate model insights and operational KPIs for stakeholder decision-making.

Junior Machine Learning Engineer

Flipkart | Hyderabad, India | Jan 2022 – May 2023

- Developed and evaluated machine learning models using Random Forest, Logistic Regression, and Gradient Boosting for customer churn prediction and product recommendation, collaborating with senior data scientists on feature engineering, model validation, and performance evaluation.
- Built Python and Apache Spark ETL pipelines to process high-volume transactional and clickstream data, improving data quality and enabling reliable datasets for analytics and machine learning workflows.
- Performed exploratory data analysis, feature engineering, and model evaluation to identify customer behavior patterns and improve predictive model performance using business-relevant evaluation metrics.
- Developed Tableau dashboards and automated reporting solutions to visualize customer engagement, sales trends, and model insights, supporting data-driven decisions across product and business teams.
- Supported deployment and monitoring of machine learning models using Docker and GCP services (BigQuery, Cloud Storage, Pub/Sub), contributing to scalable inference pipelines and production reliability.

PERSONAL PROJECT

Snapr — Multi-Agent AI Deal-Finding System (Personal Project)

[GitHub](#)

- Architected a multi-agent AI system that orchestrates specialized LLM agents using LangGraph to perform real-time deal discovery, semantic retrieval, duplicate detection, and ML-based price prediction through a centralized workflow.
- Built a production-grade RAG pipeline using OpenAI GPT, FAISS/Pinecone, embedding models, and prompt engineering to retrieve, rank, and synthesize relevant deal information, while leveraging LoRA/QLoRA fine-tuning to improve domain-specific response quality.
- Deployed the application as containerized FastAPI microservices with Docker, GitHub Actions CI/CD, and serverless execution on Modal, exposing an interactive Gradio interface for end-to-end inference.

EDUCATION

Master of Science in Computer and Information Sciences – Texas Tech University, Lubbock, TX

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate